

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 10: Circles

Previous year board questions, topic-wise: 4 questions

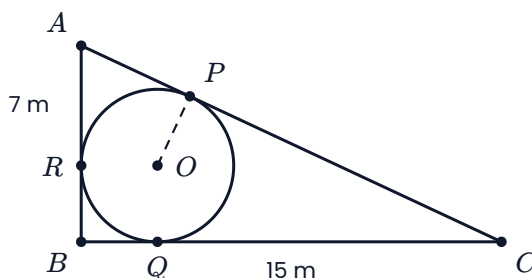
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Theorem 1: tangent and radius

- Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. **CBSE 2024, 1 mark**

2 Theorem 2: tangents from an external point

- AB is a diameter of a circle with centre O . The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90^\circ$. **CBSE 2024, 3 marks**
- A circle of radius 8 cm with centre O touches the sides AD , DC , CB and BA of a quadrilateral $ABCD$ at P , Q , R and S . $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC . **CBSE 2024, 3 marks**
- A triangular backyard ABC has a right angle at B , with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC , BC and AB at P , Q and R , and $AP = x$ m. (i) Write AR in terms of x . (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x , OR find x and then the radius r of the pit. **CBSE 2024, 4 marks, case study**



These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.